

Robotic Arm Transport

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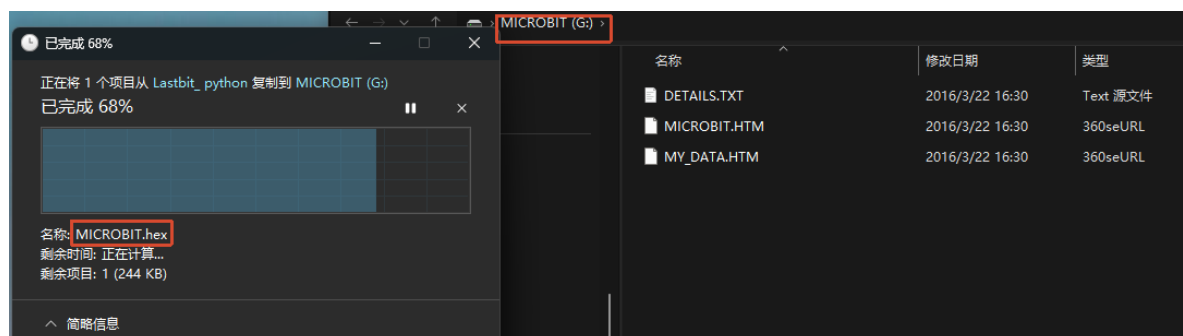
Function Description

In the previous section, we demonstrated the case of grabbing a block, where the grabbing and placing positions were the same. In this section, we introduce how to make the car transport back and forth between two positions by modifying the angles.

S1	Robotic Arm Gripper Joint
S2	Robotic Arm Vertical Joint
S3	Robotic Arm Horizontal Joint

Experiment Steps

Before flashing, make sure `LASTBIT.hex` has been flashed, otherwise an error will occur and the module cannot be imported.



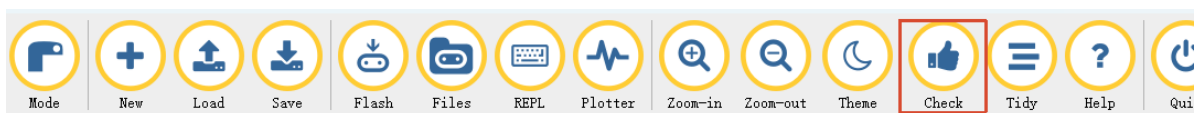
1. Before flashing, toggle the switch to the OFF position, and connect the computer to the micro:bit main board with a microUSB data cable. **Note: use the Microbit interface, not the Charging port of the expansion board.**
2. Open the Mu software and directly copy the program source code we provide. The operation steps are as follows. You can also enter the code yourself in the editor window. Note! All letters and symbols should be entered in English input mode, use the Tab key for indentation, and the last line should end with a blank program.
Click the `New` button in the toolbar at the top left corner, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether you save or not does not affect code checking and flashing.

4. At this point, you can click `Check` to check whether there are any problems with the code format.



5. At this point, if you have already connected the MICROBIT to the PC computer following the steps above and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

While flashing, the orange-yellow indicator light near the micro:bit main board will blink frequently. When the editor prompts `Copied code onto micro:bit.`, the flashing is complete, and the indicator light will stop blinking afterwards.



6. Unplug the Microusb data cable and toggle the switch to the ON position.

Code Implementation

```
# Robotic Arm Transport

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()
display.show(Image.HAPPY)
# Set all servos to 90 degrees
Lastbit.set_all_servos(90)
sleep(1000)
```

```
while True:
    # A key
    if button_a.is_pressed():
        # Drop to ready position
        Lastbit.set_servo('s2', 80)
        Lastbit.set_servo('s3', 90)
        Lastbit.set_servo('s1', 90)
        sleep(200)

        # squeeze
        Lastbit.set_servo('s1', 150)
        sleep(200)

        # lift up
        Lastbit.set_servo('s2', 160)
        sleep(200)

        # rotate
        Lastbit.set_servo('s3', 145)
        sleep(200)

        # Drop
        Lastbit.set_servo('s2', 80)
        sleep(200)

        # Release
        Lastbit.set_servo('s1', 90)
        sleep(300)

    # B key
    if button_b.is_pressed():
        # squeeze
        Lastbit.set_servo('s1', 150)
        sleep(200)

        # lift up
        Lastbit.set_servo('s2', 150)
        sleep(200)

        # rotate back
        Lastbit.set_servo('s3', 90)
        sleep(200)

        # Drop
        Lastbit.set_servo('s2', 90)
        sleep(200)

        # Release
        Lastbit.set_servo('s1', 90)
        sleep(300)

    sleep(10)
```

The source code in this section is almost identical to the previous section, except that the movement of the car from the initial grabbing position to the upper left placing position has been added. Both the starting point and the landing point of the transport can be modified by changing the servo angles.

`set_servo(id,angle)`; controls the servo to move to the `angle` angle. angle supports setting 0~180 (degrees).

id supports the following parameters:

- 'S1'
- 'S2'
- 'S3'
- 'S4'
- 'P1'
- 'P12'

Experiment Result

Place the EVA block in the center of the gripper.



Press button A to grab and lift the object, and move the servo to the front left position of the car.



Press button B to move the EVA block back from the front left position to its original position.



Notes

Make sure the car battery is sufficiently charged, otherwise there may be problems such as inability to control or insufficient motor torque.